

COURSE 3 - DATA ENGINEERING 2026

C3-062 - Distributed Processing with Spark Protected Review

Open only after the matching Pxx attempt is saved. This is repair guidance, not a solution bank.

Artifact	C3-062
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

Protected-use rule

For one attempted task only: compare your reasoning with the review, identify the failed competency, close this file, then solve a changed task. Keep original and retry evidence append-only.

R01 - Review for P01

What strong evidence shows

You distinguish partitions/tasks/stages/jobs and justify Spark only when distribution solves a real size/throughput/platform need.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you call rows "tasks" or assume every transformation launches a job, inspect the action boundary and redraw.

Changed retry

Change the example to 80 MB and argue whether local pandas could be more appropriate.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R02 - Review for P02

What strong evidence shows

You separate driver coordination from executor work, identify lazy/action boundaries and prove any driver collection is bounded.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you think `collect()` is always wrong, refine: it is acceptable only when you have proven the result is bounded and intentionally driver-sized.

Changed retry

Review a second pipeline that collects distinct customer IDs and decide safe/unsafe from an explicit upper bound.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R03 - Review for P03

What strong evidence shows

You use an explicit schema and record Spark/Python/Java/session evidence; parse failures are not silently accepted as business nulls.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If a type mismatch becomes null, separate legitimate null from parse failure instead of silently accepting it.

Changed retry

Read a second shipment fixture with an explicit schema and compare nullable rules.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R04 - Review for P04

What strong evidence shows

You use built-in Column expressions for distributed logic, keep rejects visible and preserve the declared fact grain.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you reach for a row loop, rewrite the rule as Column expressions; if impossible, document why a UDF is justified.

Changed retry

Add a changed discount rule and prove row grain remains one row per order_id.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R05 - Review for P05

What strong evidence shows

You prove dimension-key uniqueness and fact grain before choosing broadcast/shuffle from evidence.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If totals inflate, do not add DISTINCT. Inspect dimension uniqueness and join grain.

Changed retry

Use a deliberately duplicated product dimension and show how the validation blocks the join.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R06 - Review for P06

What strong evidence shows

You state grain before groupBy/window, use deterministic ordering where rank matters and reconcile values.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If a window unexpectedly changes rows, inspect joins/filters around it; window itself should not collapse rows like groupBy.

Changed retry

Create a product-category rank per region while preserving one row per order.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R07 - Review for P07

What strong evidence shows

You distinguish legitimate missing values from parse failures and prove source = clean + rejected.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If casts yield null, compare raw text and required-field policy before fillna/coalesce.

Changed retry

Add a new malformed timestamp row and ensure it is rejected with a reason rather than converted to business null.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R08 - Review for P08

What strong evidence shows

You treat CSV/JSON/Parquet/Delta as different contracts, verify Parquet outputs and do not imply Delta is bundled with PySpark.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you create one file per record or coalesce(1) by habit, reconsider output scale and downstream parallelism.

Changed retry

Write a second domain to Parquet with a different partition strategy and justify it.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R09 - Review for P09

What strong evidence shows

You prove DataFrame/SQL semantic parity instead of assuming it, including null/filter/grouping behavior.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If SQL and DataFrame differ, compare filters/null semantics/grouping columns before blaming the engine.

Changed retry

Translate a changed high-value metric both ways and compare.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R10 - Review for P10

What strong evidence shows

You connect repartition/coalesce to redistribution, tasks and output files; no default coalesce(1).

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If fewer partitions makes runtime worse, consider lost parallelism, skew and stage boundaries.

Changed retry

Choose a partition plan for a changed dataset with one hot region and four cold regions.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R11 - Review for P11

What strong evidence shows

You read formatted plans for Scan/Filter/Exchange/Join/Aggregate/Adaptive nodes and connect them to code/data.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you see a slow job but no plan evidence, capture explain/UI signals before changing configs.

Changed retry

Annotate a second plan where the dimension is small enough to consider broadcast.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R12 - Review for P12

What strong evidence shows

You cache only reused expensive intermediates, materialize intentionally and unpersist; one-pass data stays uncached.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If cache seems to make no difference, verify materialization and reuse count; tiny local jobs can hide realistic cost.

Changed retry

Re-evaluate when the reused DataFrame is 20x larger than executor memory.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R13 - Review for P13

What strong evidence shows

You measure key/partition/task imbalance, fix bad keys first and choose a proportionate skew remedy.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If you immediately salt, first prove the key is valid and that a simpler strategy cannot solve it.

Changed retry

Analyze a changed fixture where UNKNOWN contains 40% of rows and decide whether data quality is the real problem.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R14 - Review for P14

What strong evidence shows

You explain that AQE uses runtime statistics and preserves business correctness as a separate obligation.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If a "tuning" change has no hypothesis or correctness check, revert and design a controlled comparison.

Changed retry

Consider a cluster-sized dataset where initial shuffle partitions are high and AQE coalesces at runtime.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R15 - Review for P15

What strong evidence shows

You test exact transformed data plus schema/grain/rejection/totals invariants with deterministic fixtures.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If a test only checks row count, add value/grain/schema or failure-behavior evidence.

Changed retry

Add a duplicate product fixture and assert the join validation blocks it.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R16 - Review for P16

What strong evidence shows

You separate reusable transforms from CLI/config/I/O and prove a clean environment can reproduce the run.

Check your attempt

- Correct input/output grain and business invariant
- No hidden fan-out, silent rejection or fabricated runtime evidence
- Physical-plan/partition/performance claim supported by observed evidence when runtime exists
- Learner can explain the decision without reading this page

Targeted repair

If a teammate cannot rerun it from README, the phase is not finished even if your notebook works.

Changed retry

Run or dry-run the same package against a fresh shipment dataset with different paths.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.